INCH POUND

MIL-C-915/61D
29 September 1989
SUPERSEDING
MIL-C-915/61C
18 June 1985

MILITARY SPECIFICATION SHEET

CABLE, POWER ELECTRICAL, FOR OUTBOARD USE ONLY
(NOT FOR INBOARD USE), TYPE S2S

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated AWG no. 18, class D stranding. Maximum diameter 0.050 inch.
Second - Cross-linked thermosetting polyethylene insulation, colored black. Maximum wall thickness 0.060 inch.
Third - Braided shield of AWG no. 34 tin coated copper strands. Sixteen carriers, six ends per carrier, eight picks per inch.
Fourth - Synthetic rubber insulation. Nominal wall thickness 0.055 inch.
Fifth - Braided shield consisting of 0.0063-inch diameter tin-coated, copper-clad steel wires, class 40A in accordance with ASTM B 452. Twenty-four carriers, eight ends per carrier, nine or ten picks per inch.
Sixth - Two-layer jacket with the layers bonded together, with a reinforcing braid between layers, and having a total thickness of 0.065-inch minimum. Inner layer of synthetic rubber insulation having 0.033-inch nominal wall thickness. Outer layer of arctic type

AMSC N/A FSC 6145
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REQUIREMENTS - Continued

polychloroprene. Reinforcing braid of 210-denier nylon, applied with 12 carriers, 2 ends per carrier and 3 picks per inch. Completed cable overall diameter 0.480-inch minimum, 0.500-inch maximum.

Seventh - Cable surface marking required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	7.05
Voltage withstand - volts, root mean square, minimum	
Conductor to inner shield	1500
Shield to shield	1500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to inner shield	500
Shield to shield	100
Outer shield to water	10
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) (test every shipping length) - at a gauge pressure of 1000 lbf/in², for 6 hours, leakage	Zero
Capacitance - at 1 kHz, pF/ft, maximum	37 ± 1
Group B:	
Breaking strength - pounds, minimum	150
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in², minimum	1500
Elongation - percent, minimum	250
Jacket (outer layer)	
Tensile strength - lbf/in², minimum	1800
Elongation - percent, minimum	300
Set - inch, maximum	3/8
Drip - at 95 ± 1°C	Zero
Group C:	
Flammability - inches, maximum	2
Physicals (aged)	
Air pressure heat	
Insulation	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Permanence of printing (jacket) - cycles, minimum	250

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EXAMINATION AND TESTS:

	Requirements
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Shield-conformance to material coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following: Pressure cycling test in accordance with 4.7.25, except as follows:

(1) The specimen shall consist of a 10-foot length of completed cable.
(2) Instrumentation shall be connected to one of the protruding cable ends so that capacitance can be measured continuously during the test.
(3) Hydrostatic pressure in the chamber shall be raised from atmospheric pressure to a gauge pressure of 1000 lbf/in² and held for 1 hour. The specimen shall then be given 10 cycles from a gauge pressure of 1000 lb/in² to atmospheric to a gauge pressure of 1000 lbf/in², with a 10-second dwell time at each extreme. At the end of the tenth cycle, gauge pressure shall be held at 1000 lbf/in² for 1 hour.
(4) Any capacitance measurement during the test which is less than 36 pF/ft or greater than 40 pF/ft shall constitute failure.

Cable aging and compatibility (125 ± 5°C)..... No failure

UNIT ORDERING LENGTH: 500 feet (nominal).

NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1127-14)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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